BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 2

Composite Function

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1. INTRODUCTION

Unit 1 of our journey through the mathematical landscape in a Bachelor of Computer Applications (BCA) course is a deep dive into the essential elements that constitute the building blocks of advanced computational theories and practices. This unit, titled "Introduction to Functions," unfolds the complexities and simplicities of mathematical relations and functions, setting the stage for a comprehensive understanding of how data elements interact within various sets. We begin by exploring the fundamental concept of a relation, which establishes a connection between elements across different sets. This is accompanied by the introduction of pictorial diagrams, providing a visual representation of these abstract concepts, making them more accessible and easier to grasp.

As we delve deeper, the discussions pivot to the nuanced distinctions between the domain, co-domain, and the range of a relation, elucidating how each plays a pivotal role in defining the scope and output of mathematical interactions. Functions are then spotlighted as a special kind of relation, one that ensures a unique output for every input, thus laying down a foundational framework for understanding more complex operations in computer science. Through pictorial representations, we visualize functions and their components, including the domain, co-domain, and range, further solidifying our grasp of these concepts. The unit also covers an array of functions—constant, identity, polynomial, rational, modulus, signum, exponential, logarithmic, and the greater integer function—each elucidated with graphical interpretations to enhance comprehension.

Moreover, the unit explores the intricate nature of relations and functions, dissecting types of relations such as reflective, symmetric, transitive, and equivalence relations, and delving into the specifics of one-to-one, onto functions, composite functions, inverses of functions, and binary operations. This rich tapestry of mathematical concepts not only forms the cornerstone of theoretical knowledge but also serves as a practical toolkit for solving real-world problems in computing and beyond.

As we transition into Unit 2, the focus shifts towards "Composite Functions," a concept that epitomizes the essence of mathematical elegance and functional complexity. This unit is particularly crucial for students in computer applications as it mirrors the layered nature of software development and algorithm design, where functions and procedures are often

nested and combined to achieve sophisticated outcomes. Understanding composite functions equips students with the ability to dissect and construct multi-layered computational processes, enhancing their problem-solving skills and enabling them to approach programming challenges with a more analytical and systematic mindset.

The exploration of composite functions reveals how the output of one function can seamlessly become the input for another, leading to the formation of a new, more complex function. This chaining of functions is not just a mathematical exercise but a reflection of real-world processes in computer science, where data is often passed through multiple layers of processing and transformation. By mastering the principles of function composition, students can better understand the flow of data and logic within software systems, making it easier to debug, optimize, and innovate within their coding practices.

Navigating through the intricate concepts of relations and functions demands a strategic approach to learning, one that embraces patience, practice, and a deep-seated curiosity. Students are encouraged to start with the basics, ensuring a solid grasp of fundamental definitions and properties before venturing into more complex territories. Visual aids such as pictorial diagrams and graphs play a pivotal role in demystifying abstract concepts, making them more tangible and relatable.

Engaging with a variety of examples and practical exercises is key to reinforcing theoretical knowledge and gaining proficiency. Collaborative learning, through group discussions and study sessions, can offer fresh perspectives and insights, making the learning journey less daunting and more enriching. Moreover, leveraging online resources, interactive simulations, and mathematical software can provide alternative avenues for exploration and discovery, catering to different learning styles and preferences.

The study of relations and functions within the BCA curriculum is not merely an academic endeavour but a foundational step towards cultivating a robust analytical framework, critical for navigating the complexities of computer applications and the broader realm of technology.

1.1 Objectives

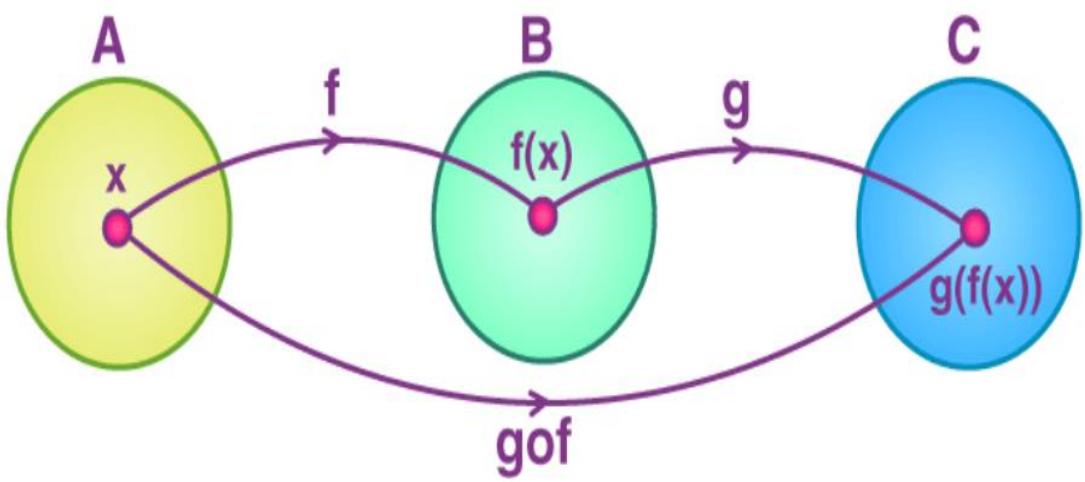
In this topic you will learn about,

- *Define the concept of composite functions and their significance in mathematics and practical applications.*
- *Explain the process of creating composite functions and interpret the results.*
- *Apply composite functions to solve problems in various contexts, such as algorithm design, data analysis, and cryptography.*
- *Analyse the properties and behaviour of composite functions through mathematical operations and real-world examples.*

2. COMPOSITE FUNCTION

Definition:

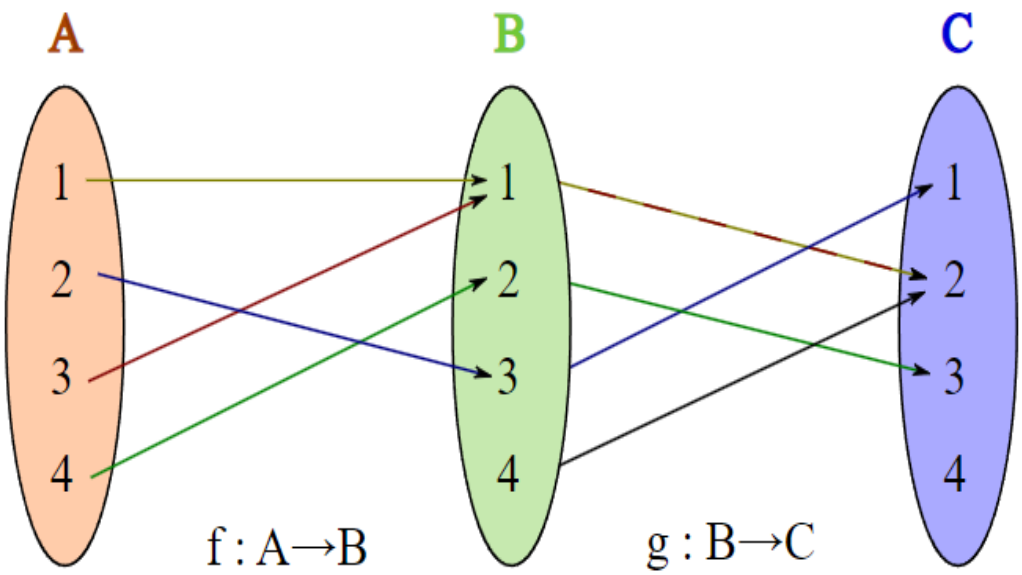
A composite function, also known as a composition of functions, is a mathematical concept that involves applying one function to the output of another function. In other words, it's the process of using the result of one function as the input for another function. This leads to the creation of a new function that encapsulates the combined behaviour of the two original functions.



Composite Function

A composite function is a fundamental concept in mathematics that arises when we combine two or more functions to create a new function.

This process involves applying one function to the output of another function. To illustrate this concept, let's consider two functions, f(x) and g(x).



Example

Given two functions, f(x) and g(x), the composite function f(g(x)) is formed by applying function g(x) to the input x and then taking the result and applying function f(x) to it. In mathematical notation, the composite function f(g(x)) can be expressed as:

f(g(x)) = f(g(x))

Here's a step-by-step breakdown of how the process works:

1. **Apply g(x) to the Input x:**
   Start by taking the input x and applying the function g(x) to it. This means substituting x into the expression for g(x) and obtaining the corresponding output. Let's say g(x) = x + 3.
   When we substitute x into this function, we get:

$$g(x) = x + 3$$

2. **Apply f(x) to the Result of g(x):**
   After obtaining the output from g(x), we take that value and apply the function f(x) to it. This results in the final value of the composite function. For instance,
   if f(x) = 2x, we have:

$$f(g(x)) = f(x + 3) = 2(x + 3) = 2x + 6$$

Properties of Composite Functions

The examination of composite functions is fundamental in the field of calculus and mathematics as a whole. When combining two functions, a new function is formed by the application of one function to the output of another. A comprehensive comprehension of the characteristics of composite functions becomes beneficial in several mathematical applications. Here are some key properties:

1. **Associativity:** Composite functions are associative, meaning that the order in which you compose functions does not affect the result. If you have three functions, f, g, and h, then $(f \circ g) \circ h$ is equivalent to $f \circ (g \circ h)$.

2. **Identity Function:** The identity function, denoted as $I(x)=x$, is a neutral element for function composition. Composing any function with the identity function results in the original function, i.e., $f \circ I = I \circ f = f$.

3. **Distributive Property:** Composition distributes over addition, subtraction, and multiplication. For example, $(f+g) \circ h = (f \circ h) + (g \circ h)$, $(f-g) \circ h = (f \circ h) - (g \circ h)$, and $(f \cdot g) \circ h = (f \circ h) \cdot (g \circ h)$.

4. **Inverse Functions:** If a function has an inverse, then the composition of the function with its inverse yields the identity function. For example, if f has an inverse f^{-1}, then $f \circ f^{-1} = I$ and $f^{-1} \circ f = I$, where I is the identity function.

5. **Domain and Range:** The domain of a composite function is determined by the domain of the innermost function, while the range is determined by the range of the outermost function.

6. **Not Commutative:** Function composition is not commutative, meaning that the order of composition matters. In general, $f \circ g$ is not equal to $g \circ f$, unless f and g are inverses of each other.

7. **Transitivity:** If you have three functions, f, g, and h, and $g \circ f$ and $h \circ g$ are defined, then you can compose them as $(h \circ g) \circ (f \circ h)$, leading to a new composite function.

8. **Invertibility:** A composite function is invertible if and only if each of the component functions is invertible. The inverse of a composite function is the composite of the inverses of the component functions in the reverse order.

Example:

Let's work through an example to solidify our understanding of composite functions:

Given functions f(x) = 2x and g(x) = x + 3, find the composite function f(g(x)).

1. Apply g(x) to the input x: g(x) = x + 3.
2. Take the result of g(x) and apply f(x): f(g(x)) = f(x + 3) = 2(x + 3) = 2x + 6.

Understanding composite functions is crucial in various fields, including computer science and data analysis. In programming, composite functions allow us to create complex processes by combining simpler functions. For instance, in data processing, a composite function might represent a sequence of data transformations. In cryptography, composite functions play a role in encryption and decryption processes.

Real-world Applications

Composite functions find relevance beyond theoretical mathematics, influencing various practical applications. In finance, they can model the effects of multiple financial transactions. In physics, they describe the successive transformations of physical quantities. By comprehending composite functions, we gain insights into real-world phenomena and their mathematical representation.

Computer Science Connection

In computer science, composite functions align with the concept of function chaining. This concept is leveraged when designing algorithms and writing code. By breaking down tasks into smaller functions and composing them, programmers build efficient and modular solutions.

Problems

Problem 1: Given functions f(x) = 2x + 1 and g(x) = x^2, find the composite function f(g(x)).

Solution: To find the composite function f(g(x)), we need to apply the function g(x) first and then apply the function f(x) to the result.

- Apply g(x) to x: g(x) = x^2.

- Apply f(x) to g(x): f(g(x)) = f(x^2) = 2(x^2) + 1 = 2x^2 + 1.

  Ans: f(g(x)) = 2x^2 + 1

Problem 2: Let h(x) = 3x - 4 and k(x) = x^3. Determine the composite function h(k(x)).

Solution: To find the composite function h(k(x)), apply function k(x) first and then function h(x) to the result.

- Apply k(x) to x: k(x) = x^3.
- Apply h(x) to k(x): h(k(x)) = h(x^3) = 3(x^3) - 4 = 3x^3 - 4.

  Ans: h(k(x)) = 3x^3 - 4

Problem 3: Consider f(x) = √x and g(x) = 2x. Find the composite function f(g(x)).

Solution: Apply function g(x) first and then function f(x) to the result to find f(g(x)).

- Apply g(x) to x: g(x) = 2x.
- Apply f(x) to g(x): f(g(x)) = f(2x) = √(2x).

  Ans: f(g(x)) = √(2x)

Problem 4: Given u(x) = x + 5 and v(x) = x^2 - 3x. Calculate the composite function v(u(x)).

Solution: To find v(u(x)), apply function u(x) first and then function v(x) to the result.

- Apply u(x) to x: u(x) = x + 5.
- Apply v(x) to u(x): v(u(x)) = v(x + 5) = (x + 5)^2 - 3(x + 5).

  Ans: v(u(x)) = x^2 + 10x + 25 - 3x - 15

Problem 5: If m(x) = 2x and n(x) = x - 3, find the composite function m(n(x)).

Solution: To find m(n(x)), apply function n(x) first and then function m(x) to the result.

- Apply n(x) to x: n(x) = x - 3.
- Apply m(x) to n(x): m(n(x)) = m(x - 3) = 2(x - 3).

  Ans: m(n(x)) = 2x - 6

Problem 6: Given f(x) = x^2 and g(x) = 2x - 1, determine the composite function g(f(3)).

Solution: Substitute 3 into f(x) and then the result into g(x) to find g(f(3)).

- Apply f(x) to 3: f(3) = 3^2 = 9.
- Apply g(x) to f(3): g(f(3)) = g(9) = 2(9) - 1 = 18 - 1 = 17.

  Ans: g(f(3)) = 17

Problem 7: Consider h(x) = √(x + 4) and k(x) = 3x. Calculate the composite function h(k(2)).

Solution: Substitute 2 into k(x) and then the result into h(x) to find h(k(2)).

- Apply k(x) to 2: k(2) = 3(2) = 6.
- Apply h(x) to k(2): h(k(2)) = h(6) = √(6 + 4) = √10.

  Ans: h(k(2)) = √10

Problem 8: Given functions f(x) = x^3 and g(x) = x + 2, find the composite function g(f(x)).

Solution: Apply function f(x) first and then function g(x) to the result to find g(f(x)).

- Apply f(x) to x: f(x) = x^3.
- Apply g(x) to f(x): g(f(x)) = g(x^3) = x^3 + 2.

  Ans: g(f(x)) = x^3 + 2

Problem 9: Let p(x) = x^2 + 1 and q(x) = 4x. Determine the composite function q(p(x)).

Solution: Apply function p(x) first and then function q(x) to the result to find q(p(x)).

- Apply p(x) to x: p(x) = x^2 + 1.
- Apply q(x) to p(x): q(p(x)) = q(x^2 + 1) = 4(x^2 + 1).

  Ans: q(p(x)) = 4x^2 + 4x + 4

Problem 10: Consider r(x) = x - 2 and s(x) = x^2. Calculate the composite function s(r(x)).

Solution: Apply function r(x) first and then function s(x) to the result to find s(r(x)).

- Apply r(x) to x: r(x) = x - 2.
- Apply s(x) to r(x): s(r(x)) = s(x - 2) = (x - 2)^2.

  Ans: s(r(x)) = x^2 - 4x + 4

3. OPERATIONS ON A FUNCTION

When dealing with functions, you can perform various operations such as addition, subtraction, multiplication, and division. These operations can be applied to functions to create new functions, often referred to as the sum, difference, product, and quotient of functions. Let's explore each of these concepts:

1. **Sum of Functions:** The sum of two functions, denoted as $(f+g)(x)$, is a new function obtained by adding the values of the two functions at the same input: $(f+g)(x)=f(x)+g(x)$.

   For example, if you have two functions $f(x)=2x$ and $g(x)=3x^2$, then the sum of these functions is $(f+g)(x)=2x+3x^2$.

2. **Difference of Functions:** The difference of two functions, denoted as $(f-g)(x)$, is a new function obtained by subtracting the values of one function from the other at the same input: $(f-g)(x)=f(x)-g(x)$.

   For instance, if you have two functions $f(x)=5x$ and $g(x)=2x^3$, then the difference of these functions is $(f-g)(x)=5x-2x^3$.

3. **Product of Functions:** The product of two functions, denoted as $(f\cdot g)(x)$, is a new function obtained by multiplying the values of the two functions at the same input: $(f\cdot g)(x)=f(x)\cdot g(x)$.

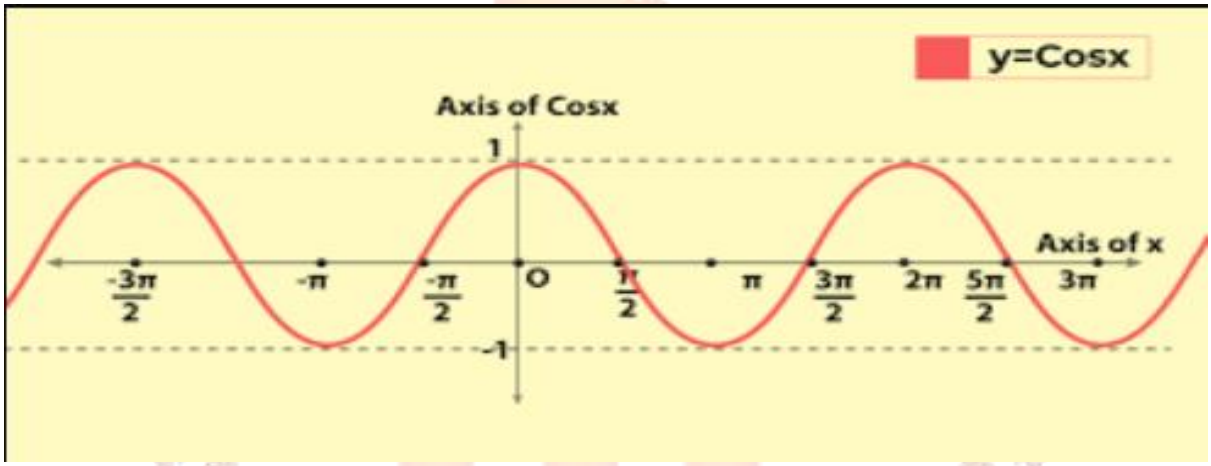
   As an example, if you have two functions $f(x)=x^2$ and $g(x)=3x$, then the product of these functions is $(f\cdot g)(x)=x^2\cdot 3x=3x^3$.

4. **Quotient of Functions:** The quotient of two functions, denoted as $(f/g)(x)$, is a new function obtained by dividing the values of one function by the values of the other at the same input: $\left(\frac{f}{g}\right)(x) = \frac{2x}{x^2} = \frac{2}{x}$.

   For instance, if you have two functions $f(x)=2x$ and $g(x)=x^2$, then the quotient of these functions is $\left(\frac{f}{g}\right)(x) = \frac{2x}{x^2} = \frac{2}{x}$.

4. PERIODIC FUNCTION

A periodic function is a type of mathematical function that repeats its values at regular intervals. These intervals are often called "periods," and the function's behaviour within each period is consistent. Periodic functions have widespread applications in various fields, including physics, engineering, and signal processing. Let's see about periodic functions:



Example Periodic Function

1. **Period:** The period of a periodic function is the length of one complete cycle or repetition of the function. Mathematically, if f(x) is a periodic function, the period is denoted as T, and it satisfies the condition:

   $f(x)=f(x+T)$

   This means that for any value of x, f(x) at x and f(x+T) at x+T are equal. The function's values repeat with a period of T.

2. **Frequency:** The frequency of a periodic function is the number of cycles or repetitions that occur within a unit interval. It is the reciprocal of the period and is often denoted as f. The relationship between frequency and period is $f=1/T$.

3. **Amplitude:** The amplitude of a periodic function is the maximum absolute value of the function within one period. It represents the vertical distance from the midline (if present) to the highest or lowest point in the function.

4. **Midline:** Some periodic functions have a midline, which is a horizontal line that represents the average or central value of the function over a period. The midline helps

identify the amplitude and understand whether the function oscillates above and below it.

Examples of Periodic Functions:

1. **Sine and Cosine Functions:** The sine (sin(x)) and cosine(cos(x)) functions are classic examples of periodic functions with a period of 2π. They repeat their values every 2π units and are used to model various waveforms, such as sound and light waves.

2. **Square Wave:** The square wave is another common periodic function. It alternates between two values, typically 1 and −1, and has a period that determines how often it switches between these values.

3. **Sawtooth Wave:** The sawtooth wave is a periodic function that rises linearly and then drops suddenly before repeating the pattern. The period of the sawtooth wave defines how long it takes to complete this rise-and-fall cycle.

4. **Triangular Wave:** The triangular wave is similar to the sawtooth wave but has a symmetric, triangular shape within each period.

Applications:

Periodic functions have diverse applications, including:

- Modeling periodic phenomena in science and engineering, such as oscillations, vibrations, and waveforms.
- Signal processing for tasks like sound and image analysis.
- Electrical engineering to describe alternating current (AC) circuits.
- Music theory and the analysis of musical sounds.
- Analyzing and understanding the behavior of physical systems, including pendulums and spring-mass systems.

In conclusion, periodic functions hold a basic position in the realm of mathematics and possess extensive applications across diverse domains. A comprehensive grasp of the characteristics of these functions, such as their period, frequency, amplitude, and midline, is crucial for effectively manipulating them and comprehending the recurring patterns they depict.

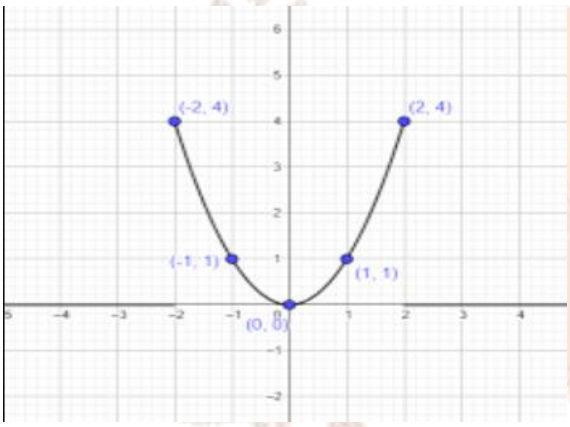
Properties of Even and Odd Function

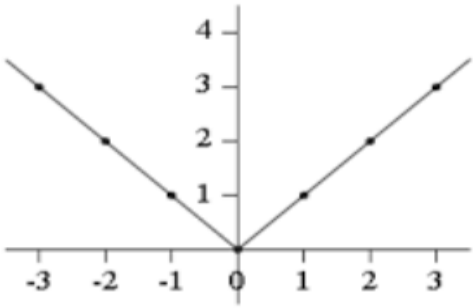
Even and odd functions are significant categories of functions in the field of mathematics, specifically within the realm of real-valued functions. The aforementioned attributes pertain to the behavioural characteristics of a function in relation to reflections across the y-axis, commonly referred to as even functions, and the origin, known as odd functions.

Even Functions:

An even function is a function f(x) that satisfies the following property:

$f(-x)=f(x)$ for all x in the domain of f.





Even Function

In other words, if you reflect the graph of an even function across the y-axis, it remains unchanged. The key properties of even functions include:

1. **Symmetry:** The graph of an even function is symmetric with respect to the y-axis. This means that if you fold the graph along the y-axis, the two halves will coincide.

2. **Even Powers:** Even functions often contain even powers of x, such as x^2, x^4, etc. However, not all functions with even powers are even functions. It's the symmetry property that defines even functions.

3. **Examples:** The most well-known even functions are $f(x)=x^2$ and $f(x)=\cos(x)$, where the cosine function has even symmetry.

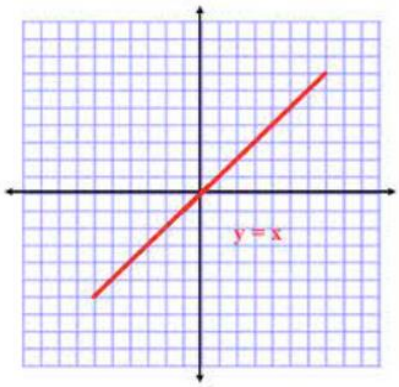
4. **Properties:** Even functions have the following properties:
   - $f(0)=0$.

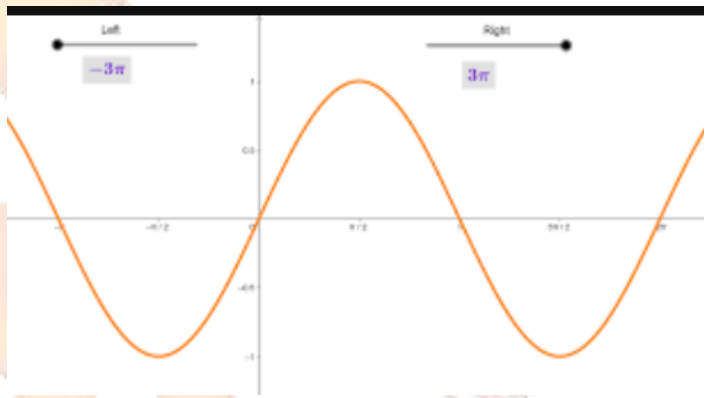
- The integral of an even function over a symmetric interval about the y-axis is zero: $\int_{-a}^{a} f(x)dx = 0$.

Odd Functions:

An odd function is a function f(x) that satisfies the following property:

f(−x)=−f(x) for all x in the domain of f.





Odd Function

In other words, if you reflect the graph of an odd function across the origin, it is flipped and negated. The key properties of odd functions include:

1. **Symmetry:** The graph of an odd function is symmetric with respect to the origin (0,0). This means that if you rotate the graph by 180 degrees about the origin, it remains unchanged.

2. **Odd Powers:** Odd functions often contain odd powers of x, such as x, x^3, x^5, etc. However, not all functions with odd powers are odd functions. It's the symmetry property that defines odd functions.

3. **Examples:** Common examples of odd functions are f(x)=x, f(x)=sin(x), and f(x)=tan(x), where the sine and tangent functions have odd symmetry.

4. **Properties:** Odd functions have the following properties:

   - f(0)=0.

   - The integral of an odd function over a symmetric interval about the origin is zero: $\int_{-a}^{a} f(x)dx = 0$

Relation between Even and Odd Functions:

It's important to note that a function can be neither even nor odd, both even and odd, or simply one of the two.

- A function can be neither even nor odd if it does not exhibit the required symmetry properties.

- A function can be both even and odd if it satisfies both properties. In this case, the function is identically equal to zero.
- A function can be either even or odd, but not both. These two properties are mutually exclusive.

5. SUMMARY

So, we've journeyed through the mathematical realms of relations, functions, and now, composite functions in our BCA curriculum. It's quite the adventure, isn't it? We kicked off with Unit 1, delving into the basics of relations and functions, painting pictures with pictorial diagrams, and understanding the critical elements like domains, co-domains, and ranges. It's like setting the stage before the actors come on – each part playing a crucial role in the grand scheme of mathematical relationships.

Then we ventured into the world of functions, a special kind of relation, where every input has its unique output. We explored various functions – constant, identity, polynomial, you name it. It's like having different tools in our toolkit, each serving a unique purpose, from the simplest of tasks to solving complex problems. And let's not forget the fascinating types of relations we encountered – reflective, symmetric, transitive, and equivalence relations, alongside one-to-one and onto functions, composite functions, and more. It's like learning the different moves in a chess game, where each move opens up new possibilities and strategies.

Now, moving onto Unit 2, we dove into composite functions, a concept that's like combining spices to create a new flavor. Here, we see how the output of one function can seamlessly flow into another, creating a new function that's a blend of both. It's like passing the baton in a relay race, where the performance of the team hinges on the seamless handoff between runners. This idea of function composition is not just a mathematical exercise but mirrors real-world processes, especially in computer science, where functions and procedures are often nested and combined to achieve sophisticated outcomes.

This journey through the mathematical foundations in our BCA course is more than just learning theories; it's about equipping ourselves with a robust analytical framework. It's fascinating how these abstract concepts find their way into practical applications, be it in

software development, algorithm design, or even in understanding the natural phenomena around us.

As we wrap up this unit, remember that mastering these concepts is akin to sharpening our tools, preparing us to tackle more complex challenges ahead. Engaging with these ideas, experimenting with examples, and collaborating with peers not only solidifies our understanding but also makes this journey an enriching experience. So, as we move forward, let's keep the curiosity alive, continue to question, and explore the vast landscape of mathematics and its applications in the world of computer applications.

6. SELF-ASSESSMENT QUESTIONS

SELF-ASSESSMENT QUESTIONS – 1

1. What is a composite function?
   a) A function that has multiple inputs and multiple outputs.
   b) A function that combines two or more functions to create a new function.
   c) A function that is defined using complex mathematical expressions.
   d) A function that only takes integers as input.
2. Which process describes the creation of a composite function?
   a) Applying one function to the output of another function.
   b) Applying both functions to the same input simultaneously.
   c) Applying a function to its own output.
   d) Combining multiple functions into a single function.
3. If f(x) = 3x and g(x) = x^2, what is the composite function f(g(x))?
   a) f(g(x)) = 3x^2
   b) f(g(x)) = 9x
   c) f(g(x)) = x^2 + 3
   d) f(g(x)) = x^3

4. What is the mathematical notation for the composite function of f(g(x))?
   a) $f \circ g(x)$
   b) $g(f(x))$
   c) $g(x) \circ f(x)$
   d) $f(x) \circ g(x)$
5. How is understanding composite functions useful in computer science and data analysis?
   a) Composite functions simplify complex mathematical problems.
   b) Composite functions only apply to abstract mathematical concepts.
   c) Composite functions help create encryption keys.
   d) Composite functions allow the creation of complex processes by combining simpler functions.
6. What is the sum of two functions $f(x)=2x^2+3x-1$ and $g(x)=4x^2-x+2$?
   a) $6x^2+2x+1$
   b) $6x^2+2x-1$
   c) $6x^2+2x$
   d) $6x^2+4x-3$
7. Which of the following represents the quotient of functions $f(x)=3x^2-5x+1$ and $g(x)=x-2$?
   a) $3x-1$
   b) $3x-1+\frac{2}{x-2}$
   c) $3x-1-\frac{2}{x-2}$
   d) $3x-1-2(x-2)$

8. Which of the following functions is an example of a periodic function?
   a) $f(x)=e^x$
   b) $f(x)=\sin(x)$
   c) $f(x)=\ln(x)$
   d) $f(x)=x^2+3x-5$
9. Which of the following properties correctly describes an odd function?
   a) $f(x)=f(-x)$
   b) $f(x)=-f(-x)$
   c) $f(x)=-f(x)$
   d) $f(x)=f(x+1)$

7. TERMINAL QUESTIONS

1. Why is understanding composite functions important in mathematical analysis and real-world applications?
2. How does the order of composition, that is, f(g(x)) vs. g(f(x)), impact the resulting composite function?
3. Given f(x) = 2x + 1 and g(x) = x^2 - 3x, find the composite function f(g(x)).
4. If h(x) = x^3 and k(x) = 2x - 5, calculate the composite function h(k(x)).
5. Consider p(x) = x^2 + 3 and q(x) = √x. Determine the composite function p(q(x)).
6. Given m(x) = 3x + 2 and n(x) = x^2 - 4x + 5, calculate the composite function m(n(x)).
7. For f(x) = 4x - 1 and g(x) = x^2 + 2x, find the composite function g(f(x)).
8. If u(x) = x^2 and v(x) = √x, determine the composite function u(v(x)).
9. Consider r(x) = x^2 + 4 and s(x) = 3x - 2. Calculate the composite function s(r(x)).
10. Given functions a(x) = x + 5 and b(x) = 2x - 3, find the composite function a(b(x)).

8. ANSWERS

SAQ Answers:

1. b) A function that combines two or more functions to create a new function.
2. a) Applying one function to the output of another function.
3. a) f(g(x)) = 3x^2
4. a) $f \circ g(x)$
5. d) Composite functions allow the creation of complex processes by combining simpler functions.
6. a) $6x^2+2x+1$
7. c) $3x-1-\frac{2}{x-2}$
8. b) f(x)=sin(x)
9. b) f(x)=−f(−x)

Terminal Question Answers:

1. Understanding composite functions helps analyse complex relationships between functions and is crucial for modelling real-world scenarios where multiple processes interact.
2. The order of composition affects the outcome of the composite function, as changing the order may yield different mathematical expressions and results.
3. f(g(x)) = 2(x^2 - 3x) + 1 = 2x^2 - 6x + 1.
4. f(g(x)) = 2(x^2 - 3x) + 1 = 2x^2 - 6x + 1.
5. p(q(x)) = (√x)^2 + 3 = x + 3.
6. m(n(x)) = 3(x^2 - 4x + 5) + 2 = 3x^2 - 12x + 15 + 2 = 3x^2 - 12x + 17.
7. g(f(x)) = (4x - 1)^2 + 2(4x - 1) = 16x^2 - 8x + 1 + 8x - 2 = 16x^2 - 1.
8. u(v(x)) = (√x)^2 = x.
9. s(r(x)) = 3(x^2 + 4) - 2 = 3x^2 + 12 - 2 = 3x^2 + 10.
10. a(b(x)) = x + 5(2x - 3) = x + 10x - 15 = 11x - 15.